

Capstone Project Report: CV Creation using LLMs

1. Introduction

1.1 Context and Background of the Project

The modern recruitment landscape is dominated by Applicant Tracking Systems (ATS), which filter 75% of resumes before human review. Job seekers face significant challenges in tailoring resumes to specific job descriptions while maintaining ATS compatibility. Manual CV customization is time-consuming and often fails to optimize keyword matching, resulting in qualified candidates being automatically rejected.

1.2 Motivation for Selecting this Topic

This project addresses the critical need for automated, intelligent CV generation that:

- Eliminates manual resume tailoring for each job application
- Ensures optimal ATS compatibility through intelligent keyword matching
- Leverages state-of-the-art Large Language Models for natural language understanding and generation
- The growing adoption of AI in recruitment necessitates equally sophisticated tools for job seekers to remain competitive in the application process.

2. Problem Statement

Job seekers struggle with three primary challenges:

- **Format Incompatibility:** Generic resumes lack job-specific keyword optimization, resulting in low ATS scores and automatic rejection
- **Manual Effort:** Tailoring resumes for each application is labor-intensive and error-prone
- **Keyword Gap:** Candidates fail to identify and integrate critical keywords from job descriptions effectively

3. Objectives

- **Automated Data Extraction:** Develop LLM-based modules to extract structured information from unstructured resume PDFs and job descriptions
- **Intelligent CV Generation:** Create an AI-driven system that generates tailored CVs by matching candidate qualifications with job requirements
- **Professional Formatting:** Generate professionally styled PDF outputs with ATS-friendly layouts
- **Interactive Refinement:** Provide optional interactive and automatic optimization modes for iterative CV enhancement

4. Methodology

4.1 Workflow / conceptual framework

The system follows a five-stage pipeline architecture:

Stage 1: PDF Text Extraction → **Stage 2:** Structured Data Extraction (Resume + JD)
 → **Stage 3:** CV Generation → **Stage 4:** CV Optimization (Optional) → **Stage 5:**
 Professional PDF Generation

4.2 Processing Flow:

- **Input:** Resume PDF + Job Description Text
- **Extraction:** Llama 3.2 model parses both inputs into structured JSON (name, contact, skills, experience, education, certifications for resume; required skills, responsibilities, keywords for JD)
- **Generation:** Gemma3 model synthesizes tailored CV by matching candidate skills with job requirements, prioritizing relevant experience and integrating JD keywords naturally into summary, skills, and experience sections
- **Optimization (Optional):** Interactive Mode: User-driven keyword addition and section enhancement. Auto-Optimize Mode: Automated keyword integration
- **Output:** Markdown-formatted CV converted to professionally styled PDF

5. System Design / Implementation

The system comprises six independent, reusable modules:

main.py : Pipeline Orchestrator, **pdf_utilities.py** : PDF I/O Layer, **resume_extractor.py** : Resume Parser, **jd_extractor.py** : Job Description Parser , **cv_creator.py** : Core CV Generator, **cv_reviewer.py** : Optimization Engine

6. Justification for Two-Stage LLM Pipeline Model Selection

The proposed two-stage LLM pipeline leverages the distinct strengths of specific compact models for resource-efficient performance: **Llama 3.2:3B** is chosen for the initial structured data extraction task, excelling in reliability and instruction following for isolating key data points. The output then feeds into **Gemma 3:1B** for the final content synthesis, which is preferred for its superior fluency and creative summarization capabilities, ensuring a high-quality narrative. This two-model strategy efficiently allocates computational resources by matching each small language model to its specialized task.

7. Conclusion and Future Work

7.1 Summary of Contributions

- End-to-End Automation: Fully automated CV generation pipeline from resume PDF to tailored CV PDF
- Intelligent Tailoring: LLM-driven content adaptation matching candidate qualifications with job requirements
- Modular Design: Reusable, maintainable architecture enabling easy extension

7.2 Future Work

- Web Interface: Develop Flask/Streamlit GUI for non-technical users
- Multimodal Integration: Analyze LinkedIn profiles, GitHub contributions for comprehensive candidate representation
- Explainable AI: Visualize which keywords/sections contribute how much to final CV